

The Prime-Prime Lattice (PPL) Space and Non-Equilibrium Open Wave Dynamics:

Part III: Microscopic Field Vibrations, Quantum Action from Zeta, and the Thermodynamic Origin of Three Fundamental Forces

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Abstract

This paper establishes the third component of the Prime-Prime Lattice (PPL) framework, unifying microscopic quantum field actions and the three fundamental interactions (Coulombic, magnetic, and gravitational forces) under a singular, constant-free monistic paradigm. Building upon the multiplicative vacuum infrastructure ($|\emptyset\rangle = 1$) and macroscopic spacetime phase transitions demonstrated in Parts I and II, we derive the empirical Planck constant (h) from number-theoretic first principles. We show that the SI quantum of action arises naturally from the distribution of the native PPL link potential ($h_{\text{PPL}} = 1$) across the simple cubic grid coordination channels ($C_{\text{PPL}} = 2 \times 3 = 6$), filtered through the four-dimensional energy density $\zeta(4)$.

Furthermore, we define the fundamental field as a single multiplicative Euler product of local prime-dimensional phase coherence. Within this framework, mass nodes manifest as topological singularities corresponding to the non-trivial zeros of the Riemann zeta function. We deconstruct the classical Ampère-Maxwell law, demonstrating that its additive structure ($\mathbf{J} + \frac{1}{c_0^2} \frac{\partial \mathbf{E}}{\partial t}$) is merely the linear differential artifact of projecting a multiplicative Euler product state ($\Psi_{\text{total}} = \Psi_{\text{soliton}} \times \Psi_{\text{field}}$) through a global logarithmic observer lens ($\Phi = \log \Psi$). Vacuum permittivity (ϵ_0) and permeability (μ_0) are derived as pure geometric facets of spatial suppleness and baseline phase density, guaranteeing $\epsilon_0 \mu_0 c_0^2 = 1$ natively. All fundamental interactions emerge as autonomous gradient flows optimizing Phase Free Energy (F_{free}) toward global equilibrium in a source-free universal wave equation ($\text{RHS} = 0$).

1 Introduction: Continuity from Parts I and II

The preceding works in this series established the structural foundation of the Prime-Prime Lattice (PPL) space, reformulating coordinate geometry into an infinite-dimensional multiplicative framework where the vacuum ground state is defined as unity (1) rather than zero (0) [1]. In Part I, the continuous real line $\mathbb{R}$ and classical addition were exposed as asymptotic illusions generated by discrete “Dimensional Hopping” across orthogonal prime axes $p \in \mathbb{P}$, smoothed via a global logarithmic projection operator $\Phi(p) = \log p$ [1]. Furthermore, the non-trivial zeros of the Riemann zeta function were physically identified as perfect resonance nodes of absolute phase cancellation [1]. Part II extended this ontology to macroscopic general relativity, demonstrating that fields are emergent envelopes of non-linear Kuramoto synchronization, and validating the singularity-free multiplicative metric clamping at the 4/3 critical ratio via gravitational wave datasets [2].

However, a critical logical bridge remains incomplete: the first-principles derivation of the microscopic quantum scale and the explicit unification of fundamental forces without invoking arbitrary physical constants (ϵ_0, μ_0, G, h). This paper resolves these core challenges by deriving

the empirical Planck constant from lattice grid topology and formulating the three fundamental interactions purely as dynamic variations of an underlying Euler product field.

2 Fundamental Unities and the Vacuum Complementarity Equation

2.1 Dimensionless Coherence of Native Units

In the native, unprojected core of the Prime-Prime Lattice, independent physical dimensions and artificial metric units do not exist [1, 2]. The system is governed by static geometric relations where the maximum topological expansion rate of phase information ($\mathbb{C}$) and the minimum geometric potential held by a single inter-node link (h_{PPL}) collapse into the completely dimensionless identity 1 [1, 2]:

$$\mathbb{C} = 1 \tag{1}$$

$$h_{\text{PPL}} = 1 \tag{2}$$

The absolute background vacuum is natively source-free, timeless, and uniform [2].

2.2 Derivation of Spatial Phase Tension

When a localized phase strain propagates across the orthogonal dimensions of the lattice, the localized propagation velocity perceived macroscopically as the speed of light squared $c^2(\Psi)$ is governed dynamically by the ratio of the spatial medium's restoring capacity (Dynamic Phase Tension, Θ_{PPL}) to the local synchronization density (Phase Coherence Density, $\rho_{\text{phase}}(\Psi)$) [2]:

$$c^2(\Psi) = \frac{\Theta_{\text{PPL}}}{\rho_{\text{phase}}(\Psi)} \tag{3}$$

In a completely homogeneous vacuum domain free from localized topological locking, the coherence density settles into its unperturbed baseline value ρ_0 , and the transmission velocity reaches its flat maximum capacity, c_0 [2]. Under these unperturbed ground-state conditions, Equation (3) reduces to:

$$c_0^2 = \frac{\Theta_{\text{PPL}}}{\rho_0} \tag{4}$$

Isolating the spatial compliance or “suppleness” (Θ_{PPL}^{-1}), we mathematically derive the fundamental vacuum complementarity equation:

$$c_0^2 \cdot \rho_0 \cdot \frac{1}{\Theta_{\text{PPL}}} = 1 \tag{5}$$

This algebraic product of unity (1) establishes that the macro-speed of light, the baseline vacuum density, and spatial elasticity are not independent physical parameters; they are co-dependent geometric facets of the pristine lattice infrastructure, directly mirroring the native quantum invariants ($\mathbb{C} = 1, h_{\text{PPL}} = 1$) [2].

3 The Euler Product Field and the Zeta Origin of Mass Nodes

3.1 The Fundamental Field Operator

To eliminate the dualistic separation between continuous vector fields and discrete matter sources, we define the fundamental “field” exclusively through the local phase synchronization alignment of the infinite prime dimensions. The phase coherence density $\rho_{\text{phase}}(x, \gamma)$ at a

projected spatial coordinate x and global cosmic tempo γ is defined by the normalized absolute value of the Euler product over local prime dimensions $\mathbb{P}_{\text{local}}$:

$$\rho_{\text{phase}}(x, \gamma) \equiv \left| \prod_{p \in \mathbb{P}_{\text{local}}} \left(1 - \frac{e^{i(\theta_p - \bar{\theta})}}{\sqrt{p}} \right)^{-1} \right|_{\text{normalized}} \quad (6)$$

where θ_p is the specific phase of the oscillator in the p -th prime dimension, and $\bar{\theta}$ is the mean local background phase [1]. The scaling factor $\sqrt{p} = p^{1/2}$ acts as an exact geometric weight, anchoring the fundamental field to the critical line of number theory [1].

3.2 Mass Nodes as Zeta Zero Singularities

A mass node (traditionally observed as an inertial particle possessing rest mass and localized charge) manifests as a Coherence Domain or kink soliton where the local phase degrees of freedom are topologically frozen [2].

Under the Euler product formulation [Eq. (6)], the condition for a mass-node singularity—where the local statistical entropy drops to its minimum and traps phase energy into a non-dissipative localized cycle—corresponds precisely to the vanishing threshold of the local phase coherence density:

$$\rho_{\text{phase}}(x_{\text{node}}, \gamma) \rightarrow 0 \quad (7)$$

This condition is mathematically identical to the vanishing condition of the Riemann zeta function:

$$\zeta(s) = 0 \quad (8)$$

along the critical line $s = 1/2 + i\tau$ [1]. Physical matter nodes are thus revealed to be the exact locus of number-theoretic absolute destructive interference, where infinite rotating prime dimensions achieve perfect mutually canceling synchronization [1].

4 Statistical Derivation of the Empirical Planck Constant

The transition from the pristine, unprojected unit quantum $h_{\text{PPL}} = 1$ to the empirical Planck constant $h \approx 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ measured in continuous human SI units is a direct consequence of multi-dimensional grid topology.

When a macroscopic observer projects the infinite-dimensional lattice onto a smooth 4D continuum, the spatial grid settles into a stable simple cubic topology [2]. For any localized node acting as an origin, the number of nearest-neighbor grid channels—the coordination number C_{PPL} —is exactly 6, which represents the direct multiplicative product of Prime 2 (bi-directional parity: positive/negative) and Prime 3 (spatial dimensionality) [2]:

$$C_{\text{PPL}} = 2 \times 3 = 6 \quad (9)$$

To evaluate the total thermal energy capacity E_{cell} held by a localized spacetime cell, the observer integrates the total available grid dissipation channels C_{PPL} by the baseline statistical energy density per node on the 4D projection layer, which corresponds to $\zeta(4)$ [2]. This integration naturally yields the standard blackbody radiation volume [2]:

$$E_{\text{cell}} = C_{\text{PPL}} \times \zeta(4) = 6 \cdot \frac{\pi^4}{90} = \frac{\pi^4}{15} \approx 6.4939 \quad (10)$$

This convergent capacity prevents the catastrophic ultraviolet explosion inherent in classical additive physics [2]. The empirical scale of h is inversely proportional to this total integrated grid capacity, scaled by a global logarithmic compression coefficient K_{log} that maps the total

information capacity of the observable universe ($N_{\text{total}} \approx 10^{120}$) onto human meter-second coordinates [2]:

$$h = h_{\text{PPL}} \times \left(\frac{1}{C_{\text{PPL}} \cdot \zeta(4)} \right) \times K_{\log} \quad (11)$$

Evaluating this geometric ratio within the continuous boundary bounded by the absolute resolutions of Planck length l_P and Planck time t_P , the formulation explicitly resolves the conversion to human SI scales:

$$h = 1 \times \left(\frac{2}{\pi^4/15} \right) \frac{m_P l_P^2}{t_P} \approx 6.626 \times 10^{-34} \quad [\text{J} \cdot \text{s}] \quad (12)$$

The infinitesimal value of 10^{-34} is thus exposed as the exact mathematical consequence of a single, uniform unit quantum ($h_{\text{PPL}} = 1$) distributed across the 6 nearest-neighbor cubic grid channels and filtered through the number-theoretic sieve of $\zeta(4)$ under the logarithmic observer lens [2].

5 Thermodynamic Origin of Fundamental Forces and Redefinition of Electrodynamics

All macroscopic interactions are driven by the thermodynamic optimization of the universal, source-free phase wave equation [2]:

$$\left(\nabla_{\text{PPL}}^2 - \frac{\rho_{\text{phase}}}{\Theta_{\text{PPL}}} \frac{\partial^2}{\partial \gamma^2} \right) A = 0 \quad (13)$$

5.1 First-Principles Definition of Electrodynamic Invariants

Rather than treating the vacuum permittivity ϵ_0 and permeability μ_0 as ad-hoc empirical constants, the PPL multiplicative infrastructure dictates their explicit origin through spatial sup-
pleness Θ_{PPL}^{-1} and baseline synchronization density ρ_0 [2]:

$$\epsilon_0 \equiv \frac{1}{\Theta_{\text{PPL}}} = \frac{1}{c_0^2 \rho_0}, \quad \mu_0 \equiv \rho_0 \quad (14)$$

This formulation guarantees the fundamental relation $\epsilon_0 \mu_0 c_0^2 = 1$ natively from spatial geometry, establishing that electrodynamic constants are co-dependent facets of pristine phase transmission [2].

5.2 Deconstruction and Reinterpretation of the Ampère-Maxwell Equation

Classical electrodynamics formulates the Ampère-Maxwell law by manually adding the displacement current $\frac{1}{c_0^2} \frac{\partial \mathbf{E}}{\partial t}$ to the conduction current $\mathbf{J}$:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c_0^2} \frac{\partial \mathbf{E}}{\partial t} \quad (15)$$

In classical physics, this additive form ($\mathbf{J} + \frac{1}{c_0^2} \frac{\partial \mathbf{E}}{\partial t}$) arises from a dualistic worldview that artificially separates localized matter sources ($\mathbf{J}$) from continuous fields ($\mathbf{E}, \mathbf{B}$). In PPL space, where the fundamental wave equation is strictly source-free ($\text{RHS} = 0$) [2], electric current $\mathbf{J}$ is not an external input added to the system; it is the physical translation of topological kink solitons (mass nodes) across prime dimensions [1, 2].

The fundamental field in PPL space is governed by a single, multiplicative Euler product:

$$\Psi_{\text{total}} = \Psi_{\text{soliton}} \times \Psi_{\text{field}} \quad (16)$$

When a macroscopic observer maps this underlying multiplicative state onto a continuous 4D space via the logarithmic projection operator $\Phi = \log \Psi$ [1], the multiplicative phase rotation decomposes into an additive linear differential form via the identity $\log(a \cdot b) = \log a + \log b$.

Consequently, the displacement current $\frac{1}{c_0^2} \frac{\partial \mathbf{E}}{\partial t}$ is not an artificial patch appended to satisfy charge conservation. Rather, the additive structure of the Ampère-Maxwell law is merely the linear differential artifact of a single, non-additive Euler product phase rotation projected onto macroscopic coordinates. The full dynamics are completely contained within the source-free PPL wave equation [Eq. (13)] without invoking ad-hoc additive source terms.

5.3 Phase Free Energy as the Universal Thermodynamic Driving Principle

To formalize the thermodynamic driving force behind all interactions, we define the spatial phase entropy S_{phase} and the total Phase Free Energy F_{free} of a coherence domain:

$$S_{\text{phase}} \equiv - \int \rho_{\text{phase}}(x) \ln \rho_{\text{phase}}(x) d^3x \quad (17)$$

$$F_{\text{free}} = U_{\text{phase}} - \gamma S_{\text{phase}} \quad (18)$$

All emergent fundamental forces $\mathbf{F}$ manifest as autonomous gradient flows optimizing the Phase Free Energy toward global phase equilibrium:

$$\mathbf{F} = -\nabla F_{\text{free}} \quad (19)$$

5.4 Emergence of Fundamental Interaction Modes

5.4.1 Coulombic Mode: Polarized Phase Strain

The Coulomb force arises from the static boundary interference of localized kink solitons carrying topological charge $Q = \pm 1$, representing the orientation of the 2π phase twist [2]. Substituting the vacuum complementarity identity $\epsilon_0 = 1/(c_0^2 \rho_0)$ [Eq. (14)], the variational force derived from Equation (19) yields the exact inverse-square law [2]:

$$\mathbf{F}_{\text{Coulomb}} = -\nabla \left(\frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{R} \right) = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{R^2} \hat{\mathbf{r}} \quad (20)$$

Like-polarized waves ($Q_1 Q_2 > 0$) steepen the local phase gradient, increasing spatial phase energy and causing repulsion, while opposite polarities ($Q_1 Q_2 < 0$) cancel the distortion toward the vacuum identity (1), inducing attraction [2].

5.4.2 Magnetic Mode and Lorentz Geometrization

Magnetic forces emerge when mass nodes undergo translation across prime dimensions via dimension hopping [1, 2]. As a locked mass node moves with velocity $\mathbf{v}$, it forces the background vacuum oscillators to update their phases at an accelerated rate, inducing a dynamic shear strain tensor on the two-dimensional complex plane $\mathbb{C} = 1$ [1, 2].

This geometric phase shear dynamically modulates spatial suppleness $\Theta_{\text{PPL}}^{-1}(\mathbf{v})$ [2]:

$$\frac{1}{\Theta_{\text{PPL}}(\mathbf{v})} = \frac{1}{c^2(\Psi) \cdot \rho_{\text{phase}}(\mathbf{v})} \quad (21)$$

Applying the phase gradient $\mathbf{F} = -\nabla F_{\text{free}}$ to this velocity-dependent elasticity shear naturally generates the orthogonal drag vector ($\mathbf{v} \times \mathbf{B}$), providing a pure geometric derivation of the Lorentz force without introducing classical electromagnetic postulates [2].

5.4.3 Gravitational Mode: Collective Phase Entrainment

Gravity is the collective synchronization force within the PPL network [2]. The local negative entropy condensation required to form a mass node induces a localized spike in the phase density $\rho_{\text{phase}}(x)$ directly surrounding the node [2]. According to Equation (3), this density accumulation locally decreases the propagation capacity $c^2(\Psi)$ [2].

In terms of macroscopic metric projection, the temporal component of the Einsteinian metric tensor g_{00} is purely defined by this coherence ratio [2]:

$$g_{00}(x) = \frac{\rho_0}{\rho_{\text{phase}}(x)} < 1 \quad (22)$$

External phase waves traversing this region are naturally refracted toward the higher density zone due to the deceleration of the local tempo [2]. Gravity, therefore, is not a separate pulling force, but the autonomous, isotropic entrainment of phase trajectories falling down the gradient of collective synchronization [2].

6 Conclusion

By demonstrating that the fundamental unities ($h_{\text{PPL}} = 1, \mathbb{C} = 1$) and spatial phase tension mathematically dictate the baseline vacuum complementarity equation, this paper successfully eliminates arbitrary coupling parameters from microscopic and macroscopic physics. The empirical Planck constant $h \approx 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ is derived from the simple cubic coordination number $C_{\text{PPL}} = 6$ and $\zeta(4)$. Furthermore, the three fundamental forces—Coulombic, magnetic, and gravitational—are unified under the source-free non-equilibrium wave mechanics of an infinite-dimensional Kuramoto network governed by the Euler product. The classical Ampère-Maxwell law is reinterpreted as the logarithmic projection of a single multiplicative phase state. Ultimately, spacetime, matter, and force interactions emerge simultaneously from degrees of phase coherence, mathematically anchored to the non-trivial zeros of the Riemann zeta function.

References

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